Conjectural permutation decoding of some AG codes

Corrected and expanded version

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Abstract

This AI-aided paper revisits the permutation-decoding conjectures of the first author (2005) for algebraic-geometric codes associated with the hyperelliptic curves $y^2 = x^p - x$. The one-point codes of length p over $\mathbb{F}_p$ turn out to be full-length Reed–Solomon codes, whose permutation group is the affine group $\mathrm{AGL}(1,p)$. This group always gives a 2-error permutation-decoding set. Exhaustive computation shows that for $p \leq 23$ it is a full permutation-decoding set for a suitable choice of information set. However, at $p = 17$ some information sets fail even though the standard counting bound is satisfied, and at $p = 37$ the counting bound rules out full permutation decoding for every information set. For the codes of length $2p(p-1)$ over $\mathbb{F}_{p^2}$ ($p \equiv 3 \pmod 4$), the stabilizer of the point at infinity was proposed as a permutation-decoding set. It acts regularly and is too small: we give an infinite family of half-rate codes for which it is not even a $(\lfloor \log_2 n \rfloor + 1)$-error decoding set. We also correct several statements about automorphism groups, separation of points, and permutation modules. In particular, we exhibit a six-dimensional Riemann–Roch module that is a submodule of a permutation module but is not itself a permutation module. A background section, a literature review, and a glossary are included.

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1 Introduction

Let X be a smooth projective curve over a finite field F, let D be a divisor on X, and let $E = P_1 + \cdots + P_n$ be a sum of distinct F-rational points not in the support of D. The algebraic-geometric (AG) code $C_X(D,E)$ is obtained by evaluating the functions in the Riemann–Roch space $L(D)$ at $P_1, \ldots, P_n$. If a group G of automorphisms of X preserves both D and E, then G permutes the coordinates of $C_X(D,E)$ and preserves the code. Permutation decoding, introduced by Prange and

MacWilliams [Pr62, Mac64], is a natural way to exploit such symmetry. A set S of code automorphisms is used to move every error pattern of bounded weight away from a fixed information set, where it can then be removed by re-encoding.

The note [Jo05], of which this paper is a corrected and expanded version, proposed that the curves

$$X: \qquad y^2 = x^p - x, \tag{1.1}$$

whose automorphism groups are unusually large, give AG codes with very efficient permutation decoding. It formulated two conjectures:

(C1) for one-point codes of length p over $\mathbb{F}_p$, permutation decoding always applies, at a cost of at most $|\mathrm{PAut}(C)| = O(p^2)$ codeword operations;

(C2) for one-point codes of length $2p(p-1)$ over $\mathbb{F}_{p^2}$, with $p \equiv 3 \pmod 4$, the image of the stabilizer of the point at infinity can serve as a full permutation-decoding set.

According to [LLMS26], these conjectures were essentially the only prior work on permutation decoding of AG codes of positive genus. This paper settles both conjectures in the negative, as universal statements, and replaces them with precise positive results, explicit obstructions, and open problems.

Summary of results.

1. The codes in (C1) are full-length Reed–Solomon codes, with permutation group $\mathrm{AGL}(1,p)$ (Theorem 7.1).

2. $\mathrm{AGL}(1,p)$ is a 2-PD-set for every information set (Theorem 7.3).

3. For $p \le 23$, $\mathrm{AGL}(1,p)$ is a full PD-set relative to some information set, in every dimension. For $p \le 13$ every information set works, but from $p = 17$ on some do not, even though the counting bound holds (Proposition 7.5, Example 7.7).

4. At $p = 37$ no information set works (Theorem 7.8). Hence (C1) fails.

5. Over $\mathbb{F}_{p^2}$, a regular group of coordinate permutations cannot correct more than about $\log n / \log(n/(n-k))$ errors by permutation decoding (Theorem 8.1). This gives an infinite family of counterexamples to (C2) (Corollary 8.2).

6. The representation of the automorphism group on $L(D)$ need not be a permutation representation, even under the strong degree hypotheses used in [Jo05] (Example 5.3).

Relation to the original note. We keep the curve-theoretic setup, the GUAVA example, and the research motivation of [Jo05]. Several statements there needed correction. They are flagged where they occur, notably in Remark 4.1 (point labels), Section 4.1 (stabilizer orders and the action of the hyperelliptic involution), Remark 5.2 (ampleness versus separation of points), and Section 5.1 (permutation modules).

Organization. Section 2 recalls the background needed later. Section 3 reviews the literature before and after 2007. Sections 4–8 contain the mathematics, Section 9 discusses complexity, and Section 10 lists open problems. The appendices contain the historical GUAVA session and a short verification script. A glossary of technical terms precedes the references.

2 Background

This section collects standard facts, with references, so that the rest of the paper can be read with modest prerequisites. Proofs are omitted.

2.1 Linear codes

A *linear code* C of length n over $F = \mathbb{F}_q$ is a subspace of F^n. The *Hamming weight* $\mathrm{wt}(v)$ of $v \in F^n$ is the number of nonzero coordinates of v. The *minimum distance* d of $C \neq 0$ is the least weight of a nonzero codeword. We say C is an $[n,k,d]$ code if $\dim C = k$. Such a code corrects any $t = \lfloor (d-1)/2 \rfloor$ errors. The Singleton bound $d \leq n-k+1$ holds, and codes attaining it are called *maximum distance separable* (MDS) [HP, MS77].

An *information set* is a set $\mathscr{I}$ of k coordinates such that projection $C \to F^{\mathscr{I}}$ is an isomorphism. The complementary set $\mathscr{J}$ of $r = n-k$ coordinates is the set of *check positions*. For an MDS code, every k-subset is an information set [MS77, Ch. 11]. After reordering coordinates so that $\mathscr{I}$ comes first, C has a *systematic* generator matrix $\mathsf{G} = (I_k \mid A)$ and parity-check matrix $\mathsf{H} = (-A^{\mathsf{T}} \mid I_r)$. The *syndrome* of $v \in F^n$ is $\mathsf{H}v^{\mathsf{T}}$.

The symmetric group S_n acts on F^n by $(\sigma v)_i = v_{\sigma^{-1}(i)}$. The *permutation automorphism group* of C is $\mathrm{PAut}(C) = \{\sigma \in S_n : \sigma(C) = C\}$. (Monomial automorphisms, which also rescale coordinates, will not be needed.)

2.2 Reed–Solomon codes and the affine group

For $1 \leq k \leq q$, the full-length Reed–Solomon code

$$\mathrm{RS}(\mathbb{F}_q, k) = \{(f(a))_{a \in \mathbb{F}_q} : \ f \in \mathbb{F}_q[x],\ \deg f < k\}$$

is a $[q, k, q-k+1]$ MDS code. The *affine group* $\mathrm{AGL}(1,q) = \{x \mapsto ux+b : \ u \in \mathbb{F}_q^{\times},\ b \in \mathbb{F}_q\}$, of order $q(q-1)$, acts on the coordinates by substitution and preserves every $\mathrm{RS}(\mathbb{F}_q, k)$. It is *sharply* 2*-transitive*: for distinct a_1, a_2 and distinct b_1, b_2 there is exactly one affine map with $a_i \mapsto b_i$.

2.3 Curves, divisors, and Riemann–Roch spaces

We follow Stichtenoth [Sti] and Hartshorne [H]. Let X/F be a smooth, projective, geometrically irreducible curve of genus g. A *divisor* is a finite formal sum $D = \sum n_P P$ of points, with degree $\deg D = \sum n_P \deg P$. For a nonzero function f, $\mathrm{div}(f) = \sum \mathrm{ord}_P(f) P$ has degree 0. The *Riemann–Roch space* is

$$L(D) = \{f \in F(X)^{\times} : \ \mathrm{div}(f) + D \geq 0\} \cup \{0\}, \qquad \ell(D) = \dim_F L(D).$$

The Riemann–Roch theorem gives $\ell(D) \geq \deg D + 1 - g$, with equality when $\deg D > 2g-2$ [Sti, Thm. I.5.17]. If $E = P_1 + \cdots + P_n$ consists of rational points outside $\operatorname{supp} D$, the evaluation map $\mathrm{ev}_E : L(D) \to F^n$, $f \mapsto (f(P_i))_i$, has image the *AG code* $C_X(D,E)$ [Go81, TVZ82]. If $\deg D < n$ then ev_E is injective, and $C_X(D,E)$ is an $[n, \ell(D), \geq n - \deg D]$ code [Sti, Thm. II.2.2]. When $D = mQ$ for a single point Q, the code is called a *one-point code*.

2.4 Hyperelliptic curves and point counts

A curve of genus $g \geq 2$ is *hyperelliptic* if it admits a degree-two map $x : X \to \mathbb{P}^1$. The associated *hyperelliptic involution* ι is central in $\mathrm{Aut}(X)$. Its fixed points are the *Weierstrass points*, which are the ramification (branch) points of x. Every automorphism of X therefore induces an automorphism of $\mathbb{P}^1$ that permutes the branch locus [BrS86, Göb]. Over $F = \mathbb{F}_q$, the *Hasse–Weil bound* says

$$\bigl|\, |X(\mathbb{F}_q)| - (q+1) \,\bigr| \leq 2g\sqrt{q}.$$

Curves attaining the upper (respectively lower) bound are called *maximal* (respectively *minimal*) [Sti, Thm. V.2.3].

2.5 Group actions and modules

A finite group G acting on a finite set Ω satisfies $|G \cdot \omega| = |G|/|G_\omega|$ (orbit–stabilizer). The action is *regular* if it is transitive with trivial stabilizers, in which case $|G| = |\Omega|$. The *permutation module* F^Ω has basis Ω permuted by G. A submodule of a permutation module need not be a permutation module. A finite-dimensional $F[G]$-module V has a composition series, and the direct sum of its composition factors is its *semisimplification*.

If $G \leq \mathrm{Aut}_F(X)$ preserves D, then G acts on $L(D)$. Throughout this paper we use the convention

$$(g \cdot f)(P) = f(g^{-1}P), \qquad \text{i.e. } g \cdot f = f \circ g^{-1}. \tag{2.1}$$

2.6 Projective linear groups

$\mathrm{PGL}_2(p) = \mathrm{GL}_2(\mathbb{F}_p)/\mathbb{F}_p^\times$ has order $p(p^2-1)$ and acts sharply 3-transitively on $\mathbb{P}^1(\mathbb{F}_p)$. Its subgroup $\mathrm{PSL}_2(p)$ of index two consists of the classes of matrices with square determinant. The stabilizer of ∞ in $\mathrm{PGL}_2(p)$ is the *Borel subgroup* $B \cong \mathrm{AGL}(1,p)$ of upper-triangular classes.

3 Literature review

3.1 Work up to 2007

Permutation decoding. Prange introduced the use of information sets to decode cyclic codes [Pr62], and MacWilliams developed permutation decoding for general systematic codes [Mac64]. A textbook account is in MacWilliams and Sloane [MS77, Ch. 16]. Early applications treated specific cyclic codes [SFT70, SF72]. Gordon gave a method for finding minimal decoding sets for the binary Golay codes [Gor82], together with the covering-type lower bound that underlies Theorem 6.3 below. Wolfmann found a set of fourteen permutations for the $[24, 12, 8]$ Golay code

[Wo83]. Chabanne extended permutation decoding to abelian codes using Gröbner bases [Ch92], and Biglieri considered group codes [Bi95]. Huffman's survey chapter [Hu98] and the textbook [HP] summarize the theory.

The notion of *partial* permutation decoding, using s-PD-sets with $s < t$, was introduced by Key, McDonough and Mavron [KMM05] and also by Kroll and Vincenti [KV05]. Permutation decoding and its partial version were applied to codes from designs, finite geometries and graphs [KMR04, KMM06, KS07]. These works emphasize two points that are central here: the choice of information set matters, and a large permutation group does not by itself guarantee full permutation decoding.

AG codes and their automorphisms. AG codes were introduced by Goppa [Go81]. The Tsfasman–Vlăduţ–Zink theorem [TVZ82] showed that they can beat the Gilbert–Varshamov bound. Standard references are [Sti, TV, HvLP]. The relation between curve automorphisms and code automorphisms was studied by Stichtenoth [St90] and by Wesemeyer [W] for plane curves. Joyner and Ksir [JK06] showed that, in many cases, $\mathrm{Aut}(X)$ and the permutation group of a corresponding AG code coincide. For genus zero, the automorphism groups of (generalized, doubly extended) Reed–Solomon codes were determined by Dür [Dü87]. The automorphism groups of generalized Reed–Muller codes over prime fields, which include the full-length Reed–Solomon codes over $\mathbb{F}_p$, were determined by Knörr and Willems [KW] and by Berger and Charpin [BC93]. Automorphism groups of hyperelliptic curves were classified by Brandt and Stichtenoth [BrS86], and computed algorithmically by Göb [Göb].

Representations on Riemann–Roch spaces. The action of $\mathrm{Aut}(X)$ on $L(D)$ was studied by Joyner and Traves [JT] and by Joyner and Ksir [JK2], building on modular representation theory such as [S]. The conjectures on permutation decoding of codes from $y^2 = x^p - x$ first appeared in [JT] and were formulated in [Jo05]. GUAVA 2.0 implemented permutation decoding [GUAVA]. Shaska and Wang [ShW07] studied automorphism groups of AG codes from $C_{a,b}$ curves, motivated in part by the decoding suggestion of Joyner and Ksir.

Decoding AG codes. Algebraic decoding of AG codes goes back to Justesen et al. [JLJHH89], Skorobogatov and Vlăduţ [SV90], and Feng and Rao [FR93]. List decoding was developed by Sudan [Su97], Guruswami and Sudan [GS99], and Shokrollahi and Wasserman [SW99]. The survey [HvLP] describes the state of the art around 1998. These algorithms decode up to (or beyond) half the designed distance in polynomial time. The interest of permutation decoding lies in its simplicity: it uses only syndrome computations and coordinate permutations.

3.2 Work after 2007

Partial permutation decoding. Partial PD-sets were constructed for codes from lattice graphs [KS08], for first-order Reed–Muller codes [Sen09], and for codes from affine geometry designs [KMM08]. Key, McDonough and Mavron further studied Reed–Muller codes [KMM10, KMM17]. Lim, Fossorier and Kavčić used code automorphisms for permutation decoding of binary images of Reed–Solomon codes [LFK10]. Bernal and Simón developed a systematic theory of information

sets for abelian codes in terms of defining sets [BS11, BS14], and used it for partial permutation decoding of abelian and affine-invariant codes [BS13, BS23].

Permutation groups of Reed–Solomon codes. Camps-Moreno, Lau, López and Santos [CLLS] gave an elementary proof that the permutation group of a Reed–Solomon code on an arbitrary evaluation set consists of the affine maps preserving that set, away from exceptional dimensions. This recovers the case of the full field used in Theorem 7.1. Joyner, Ksir and Traves [JKT] revisited automorphism groups of AG codes on $\mathbb{P}^1$ using algebraic-geometric methods.

Permutation decoding of AG codes of positive genus. Recent work has returned to the question raised in [Jo05]. Lichtenwalner, López, Matthews and Seneviratne [LLMS26] use automorphisms of Hermitian and norm-trace curves to construct PD-sets that correct specific *burst* errors for one-point codes. They note that little progress had been made for positive genus beyond the conjectures of [Jo05]. Castellanos, Tizziotti and Olaya-León [CTO26] treat curves defined by separated polynomials $f(y) = g(x)$ with f additive, including Hermitian and generalized Hermitian curves. They obtain PD-sets for bursts on points sharing a coordinate, and partial 2-PD-sets formed from two families of automorphisms.

Note that (1.1) can be written $x^p - x = y^2$, which has this separated form with the additive polynomial in x. Its Weierstrass semigroup at P_∞ is generated by 2 and p, the degrees of the two sides. The 2-PD result of Theorem 7.3 below is analogous to their two-family construction: the translations $x \mapsto x + b$ are exactly the automorphisms arising from the roots of the additive polynomial $x^p - x$. Whether the burst-error PD-sets of [CTO26, LLMS26] apply to the codes of Section 8 is raised in Problem 10.4.

4 The curve and its automorphisms

Throughout, $p \geq 5$ is prime, $g = (p-1)/2 \geq 2$, and X is the smooth projective model of (1.1). (For $p = 3$ the curve has genus one and a different automorphism group.) Since $x^p - x$ is squarefree, X is hyperelliptic of genus g. The hyperelliptic map $x : X \to \mathbb{P}^1$ has branch locus $\mathbb{P}^1(\mathbb{F}_p)$, and a unique point P_∞ lies over ∞. The finite branch points are $P_a = (a, 0)$ for $a \in \mathbb{F}_p$.

Remark 4.1 (Point labels in [Jo05]). The note [Jo05] lists $X(\mathbb{F}_p) = \{(1:0:0), (0:0:1), (1:0:1), \ldots, (p-1:0:1)\}$, so $(a:0:1)$ denotes the affine point P_a and $(1:0:0)$ denotes the point at infinity. It then puts $P_1 = (1:0:1)$ and states that the stabilizer of P_1 is generated by $(x,y) \mapsto (x,-y)$, (a^2x, ay) and $(x+1, y)$. The translation $x \mapsto x+1$ does not fix $(1,0)$, but these maps do generate the stabilizer of P_∞. We therefore read P_1 in [Jo05] as P_∞ throughout, and likewise read the point $(1:0:0)$ in conjecture (C2). (In the plane projective closure of (1.1) the point at infinity is $[0:1:0]$; divisor computations use the smooth model.)

Lemma 4.2. $\mathrm{ord}_{P_\infty}(x) = -2$ *and* $\mathrm{ord}_{P_\infty}(y) = -p$. *For every* $m \geq 0$ *the set*

$$\{x^i : \ 2i \leq m\} \ \cup \ \{x^i y : \ 2i + p \leq m\} \tag{4.1}$$

is an F*-basis of* $L(mP_\infty)$. *In particular* $\ell(mP_\infty)$ *is the number of non-gaps* $\leq m$ *of the numerical semigroup* $\langle 2, p \rangle$.

Proof. The map x has degree two and is totally ramified at P_∞, so $\mathrm{ord}_{P_\infty}(x) = -2$. The equation $y^2 = x^p - x$ then gives $\mathrm{ord}_{P_\infty}(y) = -p$. The affine curve $y^2 = x^p - x$ is smooth, because $x^p - x$ is squarefree and the characteristic is odd. Hence its coordinate ring $F[x,y]/(y^2 - x^p + x)$ is integrally closed and equals the ring of functions regular away from P_∞. Each element has a unique expression $A(x) + yB(x)$. The monomials in (4.1) have distinct pole orders (even for x^i, odd for $x^i y$), so leading terms cannot cancel. It follows that $A + yB \in L(mP_\infty)$ exactly when it is a combination of these monomials. Compare [Sti, Prop. VI.4.1]. □

Definition 4.3. A finite group $G \leq \mathrm{Aut}_F(X)$ is *large* (over F) if $|G| > |X(F)|$.

Lemma 4.4. *If G is large, every F-rational point is ramified in $X \to X/G$.*

Proof. A point $P \in X(F)$ with trivial stabilizer would have an orbit of size $|G| > |X(F)|$, which is impossible. A nontrivial stabilizer is the inertia group of P in the Galois cover $X \to X/G$. □

4.1 Automorphisms over $\mathbb{F}_p$

Let $F = \mathbb{F}_p$. Then $X(\mathbb{F}_p) = \{P_\infty\} \cup \{P_a : a \in \mathbb{F}_p\}$ has $p+1$ points, because $x^p - x = 0$ for $x \in \mathbb{F}_p$. For $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{GL}_2(\mathbb{F}_p)$ put $x' = (ax+b)/(cx+d)$. Since the entries lie in $\mathbb{F}_p$,

$$x'^p - x' = \frac{(ax^p+b)(cx+d) - (ax+b)(cx^p+d)}{(cx+d)^{p+1}} = \frac{(\det M)\,(x^p - x)}{(cx+d)^{p+1}}. \tag{4.2}$$

Thus any lift of $x \mapsto x'$ to X has the form

$$(x,y) \longmapsto \Big(\frac{ax+b}{cx+d},\ \frac{\eta\, y}{(cx+d)^{(p+1)/2}}\Big), \qquad \eta^2 = \det M. \tag{4.3}$$

Rescaling M by $\lambda \in \mathbb{F}_p^\times$ multiplies $\det M$ by λ^2, so the square class of $\det M$ is well defined in $\mathrm{PGL}_2(p)$. Hence *the projectivity lifts to an $\mathbb{F}_p$-automorphism if and only if* $\det M$ *is a square in* $\mathbb{F}_p^\times$, that is, if and only if its class lies in $\mathrm{PSL}_2(p)$. Each such class has exactly two lifts, which differ by ι.

Conversely, every automorphism of X (over $\overline{\mathbb{F}}_p$) normalizes ι and induces a Möbius transformation preserving the branch locus $\mathbb{P}^1(\mathbb{F}_p)$. A Möbius transformation is determined by the images of $0, 1, \infty$, so it lies in $\mathrm{PGL}_2(p)$. Therefore the group $G_p = \mathrm{Aut}_{\mathbb{F}_p}(X)$ fits into an exact sequence

$$1 \to \langle \iota \rangle \to G_p \to \mathrm{PSL}_2(p) \to 1, \qquad |G_p| = p(p^2-1),$$

in agreement with [BrS86, Göb]. Explicit elements are

$$\iota : (x,y) \mapsto (x,-y), \quad \delta_a : (x,y) \mapsto (a^2 x, ay), \quad \tau_b : (x,y) \mapsto (x+b, y), \quad \omega : (x,y) \mapsto \big(-1/x,\ y/x^{(p+1)/2}\big), \tag{4.4}$$

for $a \in \mathbb{F}_p^\times$ and $b \in \mathbb{F}_p$. Note that $\delta_{-1} = \iota$. The group G_p is transitive on $X(\mathbb{F}_p)$.

The stabilizer $H_p = \mathrm{Stab}_{G_p}(P_\infty)$ is generated by the δ_a and τ_b and has order

$$|H_p| = |G_p|/(p+1) = p(p-1). \tag{4.5}$$

(The value $2p(p-1)$ given in [Jo05] is the order of the corresponding stabilizer over $\mathbb{F}_{p^2}$; see Section 4.2.) Since ι fixes every branch point, the group of permutations that H_p induces on $\{P_a\}$ is

$$\{x \mapsto ux+b:\ u \in (\mathbb{F}_p^\times)^2,\ b \in \mathbb{F}_p\}, \qquad \text{of order } p(p-1)/2. \tag{4.6}$$

A nonsquare multiplier u cannot be lifted to an $\mathbb{F}_p$-automorphism fixing P_∞, by the criterion above.

Proposition 4.5. *For every $m \geq 0$, the H_p-module $L(mP_\infty)$ has a composition series with one-dimensional factors. Hence its semisimplification is a direct sum of characters of H_p.*

Proof. Order the basis (4.1) by increasing pole order at P_∞. Since H_p fixes P_∞, each subspace of functions with pole order at most j is H_p-stable. The pole orders are distinct, so the successive quotients are one-dimensional. Explicitly, with $\varepsilon \in \{0,1\}$,

$$\delta_a \cdot (x^i y^\varepsilon) = a^{-2i-\varepsilon} x^i y^\varepsilon, \qquad \tau_b \cdot (x^i y^\varepsilon) = (x-b)^i y^\varepsilon, \qquad \iota \cdot (x^i y^\varepsilon) = (-1)^\varepsilon x^i y^\varepsilon,$$

using the convention (2.1). Note that ι acts nontrivially on the basis elements involving y; the corresponding statement in [Jo05] is incorrect. □

4.2 Automorphisms and rational points over $\mathbb{F}_{p^2}$

Let $F = \mathbb{F}_{p^2}$. Every element of $\mathbb{F}_p^\times$ is a square in F, so every class in $\mathrm{PGL}_2(p)$ lifts by (4.3), and by the argument of Section 4.1 these are all the automorphisms:

$$1 \to \langle \iota \rangle \to G_{p^2} \to \mathrm{PGL}_2(p) \to 1, \qquad |G_{p^2}| = 2p(p^2-1). \tag{4.7}$$

The stabilizer $H = \mathrm{Stab}_{G_{p^2}}(P_\infty)$ is the preimage of the Borel subgroup and has order $2p(p-1)$. It consists of the maps

$$(x,y) \longmapsto (ux+b,\ \eta y), \qquad u \in \mathbb{F}_p^\times,\ b \in \mathbb{F}_p,\ \eta \in \mathbb{F}_{p^2},\ \eta^2 = u. \tag{4.8}$$

Proposition 4.6. *Suppose $p \equiv 3 \pmod 4$. The G_{p^2}-orbits on $X(\mathbb{F}_{p^2})$ are*

$$O_1 = X(\mathbb{F}_p), \quad |O_1| = p+1; \qquad O_2 = X(\mathbb{F}_{p^2}) \setminus X(\mathbb{F}_p), \quad |O_2| = 2p(p-1).$$

The subgroup H acts regularly on O_2. Consequently the stabilizer H_2 of a point $P_2 \in O_2$ has order $p+1$, $H \cap H_2 = \{1\}$, and $G_{p^2} = H \cdot H_2$. If $p \equiv 1 \pmod 4$, then $X(\mathbb{F}_{p^2}) = X(\mathbb{F}_p)$ is a single orbit of size $p+1$.

Proof. For $x \in \mathbb{F}_p$ we get $y = 0$. For $x \in \mathbb{F}_{p^2} \setminus \mathbb{F}_p$, put $z = x^p - x \neq 0$. Then $z^p = x - x^p = -z$, so $z^{p-1} = -1$ and

$$z^{(p^2-1)/2} = (z^{p-1})^{(p+1)/2} = (-1)^{(p+1)/2}.$$

Hence z is a nonzero square in $\mathbb{F}_{p^2}$ exactly when $p \equiv 3 \pmod 4$. In that case each of the p^2-p values $x \notin \mathbb{F}_p$ has two points above it, and otherwise none.

Write $x \notin \mathbb{F}_p$ uniquely as $\alpha + \beta\theta$, with $\alpha \in \mathbb{F}_p$, $\beta \in \mathbb{F}_p^\times$, for a fixed $\theta \in \mathbb{F}_{p^2} \setminus \mathbb{F}_p$. The maps $x \mapsto ux+b$ with $u \in \mathbb{F}_p^\times$, $b \in \mathbb{F}_p$ send (α,β) to $(u\alpha+b, u\beta)$, so they act regularly on these $p(p-1)$ values. For a fixed affine map the two lifts in (4.8), with $\pm\eta$, send a given point to the two points over the image value. Hence H is transitive on O_2, and since $|H| = |O_2|$ it acts regularly. The set $X(\mathbb{F}_p)$ is the fixed locus of ι together with P_∞, so it is G_{p^2}-stable, and it is a single orbit by Section 4.1. The remaining assertions follow from orbit–stabilizer. □

Remark 4.7. This agrees with the Hasse–Weil bound. For $p \equiv 3 \pmod 4$, $|X(\mathbb{F}_{p^2})| = 2p^2 - p + 1 = p^2 + 1 + 2g\,p$, so X is *maximal* over $\mathbb{F}_{p^2}$. For $p \equiv 1 \pmod 4$, $|X(\mathbb{F}_{p^2})| = p + 1 = p^2 + 1 - 2g\,p$, so X is *minimal*. The orbit statement was originally communicated by R. Guralnick [Gu]; the proof above is self-contained.

For $p \equiv 3 \pmod 4$, the invariant divisor $D_2 = \sum_{P \in O_2} P$ has degree $2p(p-1) > 2g-2$, so $\ell(D_2) = 2p(p-1) + 1 - g$.

5 AG codes and group representations

Let $G \leq \mathrm{Aut}_F(X)$ be finite, and let D and $E = \sum_{P \in \Omega} P$ be G-invariant with disjoint supports, where Ω is a set of rational points. For $g \in G$ let $\pi_E(g) \in S_\Omega$ be the coordinate permutation $(\pi_E(g)c)_P = c_{g^{-1}P}$. With (2.1),

$$\mathrm{ev}_E(g \cdot f) = \pi_E(g)\,\mathrm{ev}_E(f) \qquad (f \in L(D)). \tag{5.1}$$

Hence $\pi_E(G) \leq \mathrm{PAut}(C_X(D,E))$.

Proposition 5.1. *Write $C = C_X(D,E)$, and let $\rho_C : G \to \mathrm{GL}(C)$ be the restriction of π_E to C.*

1. $\ker \pi_E$ *is the pointwise stabilizer of Ω. This holds by the definition of a permutation action and needs no hypothesis on $L(D)$.*

2. $\ker \pi_E \leq \ker \rho_C$*, with equality if $L(D)$ separates the points of Ω.*

3. *If* ev_E *is injective (for example if* $\deg D < n$*), then* ev_E *is an isomorphism of $F[G]$-modules $L(D) \cong C$. Since C is a submodule of the permutation module F^Ω, so is $L(D)$, up to isomorphism.*

Proof. (1) is the definition. For (2), if $g \in \ker \rho_C$ then $f(g^{-1}P) = f(P)$ for all $f \in L(D)$ and $P \in \Omega$, and separation forces $g^{-1}P = P$. (3) follows from (5.1). □

Remark 5.2. The statement in [Jo05] that "D ample implies $L(D)$ separates points" requires correction. On a curve every divisor of positive degree is ample, but its complete linear system need not separate points. For example, if $g \geq 1$ then $L(P_\infty) = F$ separates no pair of points. A standard sufficient condition is $\deg D \geq 2g+1$, which makes D very ample [H, Cor. IV.3.2]. It is also known that a nontrivial automorphism of a curve of genus $g \geq 2$ has at most $2g+2$ fixed points, so π_E is faithful when $n > 2g+2$ [Sti, §VII.3].

5.1 Invariant submodules need not be permutation modules

The note [Jo05] concluded, from the result of Wesemeyer [W] under hypotheses such as $n > \max(2\deg D, 2g+2)$, that the representation of G on $L(D)$ is equivalent to one by permutation matrices. Proposition 5.1(3) only gives a *submodule* of a permutation module, and the stronger conclusion is false.

Example 5.3 (A Riemann–Roch module that is not a permutation module)**.** Let $p = 7$, $F = \mathbb{F}_{49}$, $g = 3$, and let O_1, O_2 be as in Proposition 4.6. Put $D = \sum_{P \in O_1} P$ and $E = \sum_{P \in O_2} P$. These are G_{49}-invariant with disjoint supports, and

$$\deg D = 8, \qquad n = 84 > \max\{16, 8\}.$$

Riemann–Roch gives $\ell(D) = 8 + 1 - 3 = 6$, and ev_E is injective. We have $\mathrm{div}(y) = \sum_a P_a - 7P_\infty$ and $\mathrm{div}(x) = 2P_0 - 2P_\infty$, so

$$\mathrm{div}(1/y) + D = 8P_\infty \geq 0, \qquad \mathrm{div}(x/y) + D = 2P_0 + 6P_\infty \geq 0.$$

Hence $1/y$ and x/y lie in $L(D)$. The automorphism τ_1 has order 7, and by (2.1)

$$\tau_1 \cdot (x/y) = \frac{x-1}{y} = \frac{x}{y} - \frac{1}{y} \neq \frac{x}{y}.$$

So τ_1 acts nontrivially on the six-dimensional space $L(D)$. A permutation of six letters whose order divides 7 is the identity. So no representation of G_{49} by 6×6 permutation matrices can be equivalent to the representation on $L(D)$, although all the numerical hypotheses of [Jo05] hold.

Remark 5.4. If ev_E is not injective, C is the quotient $L(D)/\ker \mathrm{ev}_E$. The decomposition of these modular representations into indecomposables, when p divides $|G|$, remains an interesting problem; see [JK2, JT, S].

6 Permutation decoding and a counting obstruction

Let C be an $[n,k,d]$ code with $t = \lfloor (d-1)/2 \rfloor \geq 1$, an information set $\mathscr{I}$ and check set $\mathscr{J}$ ($|\mathscr{J}| = r = n-k$), and systematic matrices $\mathsf{G} = (I_k \mid A)$, $\mathsf{H} = (-A^{\mathsf{T}} \mid I_r)$.

Lemma 6.1 (Syndrome criterion)**.** *Let $v = c + e$ with $c \in C$ and $\mathrm{wt}(e) \leq t$. The information coordinates of v agree with those of c if and only if $\mathrm{wt}(\mathsf{H}v^{\mathsf{T}}) \leq t$.*

Proof. Write $e = (u, e_{\mathscr{J}})$ and $s = \mathsf{H}v^{\mathsf{T}} = \mathsf{H}e^{\mathsf{T}} = e_{\mathscr{J}} - A^{\mathsf{T}}u$. If $u = 0$ then $s = e_{\mathscr{J}}$ has weight at most t. Conversely, $(u, A^{\mathsf{T}}u) \in C$ and $(u, A^{\mathsf{T}}u) = e - (0, s)$. This codeword has weight at most $\mathrm{wt}(e) + \mathrm{wt}(s) \leq 2t < d$, so it is zero and $u = 0$. □

Definition 6.2. For $1 \leq s \leq t$, a subset $S \subseteq \mathrm{PAut}(C)$ is an *s-PD-set* relative to $\mathscr{I}$ if for every set T of at most s coordinates some $\sigma \in S$ satisfies $\sigma(T) \subseteq \mathscr{J}$. A t-PD-set is a *full PD-set*.

Algorithm. Given v and an s-PD-set S, compute $\mathrm{wt}(\mathsf{H}(\sigma v)^{\mathsf{T}})$ for $\sigma \in S$ until a value $\leq t$ is found. Then re-encode the information symbols of σv to get $c_\sigma \in C$, and output $\sigma^{-1} c_\sigma$. By Lemma 6.1 this decodes correctly whenever $\mathrm{wt}(e) \leq s$. (Any σ that passes the test gives the correct answer, since $\mathrm{wt}(\sigma e) = \mathrm{wt}(e) \leq t$.)

Theorem 6.3 (Counting bound)**.** *If S is an s-PD-set relative to $\mathscr{I}$, then*

$$|S| \binom{n-k}{s} \;\geq\; \binom{n}{s}. \tag{6.1}$$

Hence if $K \leq \mathrm{PAut}(C)$ satisfies $|K| \binom{n-k}{s} < \binom{n}{s}$, then no subset of K is an s-PD-set for any information set.

Proof. A fixed σ sends exactly $\binom{r}{s}$ of the s-subsets into $\mathscr{J}$, namely the s-subsets of $\sigma^{-1}(\mathscr{J})$. All $\binom{n}{s}$ subsets must be covered. □

Remark 6.4. Inequality (6.1) is the simplest case of Gordon's covering argument [Gor82]. The iterated (Schönheim-type) version, used for example in [KMM05], is sharper. It is only a *necessary* condition; Example 7.7 shows that it is not sufficient.

7 Codes over $\mathbb{F}_p$

Let $E = \sum_{a\in\mathbb{F}_p} P_a$ and $D = mP_\infty$ with $m \geq 0$.

Theorem 7.1. *With* $k = \min\bigl(p, \lfloor m/2\rfloor + 1\bigr)$,

$$C_X(mP_\infty, E) = \mathrm{RS}(\mathbb{F}_p, k),$$

a $[p, k, p-k+1]$ *code. For* $2 \leq k \leq p-2$ *its full permutation group is* $\mathrm{AGL}(1,p)$*, of order* $p(p-1)$.

Proof. Every basis monomial in (4.1) involving y vanishes at the points P_a. So the code is spanned by the evaluations of $1, x, \ldots, x^{\lfloor m/2\rfloor}$ on $\mathbb{F}_p$. Since $x^p \equiv x$ on $\mathbb{F}_p$, its dimension is k. The permutation group is the one-variable case of the theorem of Knörr and Willems on generalized Reed–Muller codes over prime fields [KW], whose exceptional cases correspond to $k \in \{1, p-1, p\}$; see also [CLLS, BC93]. □

Remark 7.2. When $m \geq p$, the function $y \in L(mP_\infty)$ evaluates to zero on E, so ev_E is not injective. The code group $\mathrm{AGL}(1,p)$ is twice as large as the geometric image (4.6). So the full code symmetry is *not* induced by $\mathrm{Aut}_{\mathbb{F}_p}(X)$, and the curve structure adds nothing beyond the Reed–Solomon structure for these codes.

Theorem 7.3 (Two errors)**.** *Let* $1 \leq k \leq p-4$. *Then* $\mathrm{AGL}(1,p) \leq \mathrm{PAut}(\mathrm{RS}(\mathbb{F}_p, k))$ *is a* 2*-PD-set relative to every information set. More precisely, for each set* T *of two coordinates exactly* $r(r-1)$ *affine maps send* T *into* $\mathscr{J}$*, where* $r = p-k$.

Proof. Since $d = p-k+1 \geq 5$, we have $t \geq 2$. Every k-set is an information set because the code is MDS. Given $T = \{a_1, a_2\}$ and any ordered pair (b_1, b_2) of distinct check positions, sharp 2-transitivity gives a unique affine map with $a_i \mapsto b_i$. There are $r(r-1)$ such pairs. Single errors are handled by transitivity. □

Example 7.4 (The $[7,3,5]$. *code of* [Jo05]] *For* p=7 *and* D=4P$_\infty$, $L(D) = \langle 1, x, x^2\rangle$ and $C = \mathrm{RS}(\mathbb{F}_7, 3)$. The systematic matrix computed in GUAVA [Jo05] is

$$\mathsf{G} = \begin{pmatrix} 1 & 0 & 0 & 6 & 4 & 6 & 4 \\ 0 & 1 & 0 & 6 & 3 & 3 & 1 \\ 0 & 0 & 1 & 3 & 1 & 6 & 3 \end{pmatrix}.$$

We checked by direct computation that this code has minimum weight 5, that the permutations $(1,5,4)(2,3,7)$ and $(1,3,2,6,7,4)$ preserve it and generate a group of order 42, and that a search of all 5040 elements of S_7 finds no further permutation automorphisms. Thus $\mathrm{PAut}(C) \cong \mathrm{AGL}(1,7)$. Here $t = 2$, so by Theorem 7.3 the whole group is a full PD-set, and $4 \cdot 3 = 12$ of its elements handle each pair of error positions. (The note [Jo05] proposed $\mathrm{PAut}(C) \setminus \{1\}$, which also works.)

7.1 Exhaustive computations for small p

For $\mathrm{RS}(\mathbb{F}_p, k)$, every k-subset of $\mathbb{F}_p$ is an information set, and whether $\mathrm{AGL}(1,p)$ is a full PD-set depends only on the $\mathrm{AGL}(1,p)$-orbit of the information set. We tested every orbit.

Proposition 7.5 (computer verified)**.** *Let $5 \le p \le 23$ be prime and $2 \le k \le p-2$, and let $t = \lfloor (p-k)/2 \rfloor$.*

1. *For every such (p,k) there is an information set relative to which $\mathrm{AGL}(1,p)$ is a full PD-set for $\mathrm{RS}(\mathbb{F}_p, k)$.*
2. *For $p \le 13$, every information set has this property.*
3. *For $p \in \{17, 19, 23\}$, the information sets for which $\mathrm{AGL}(1,p)$ fails to be a full PD-set occur exactly for the following (p,k):*

(p,k)	t	*failing orbits / all orbits*	*failing information sets / all*
$(17,7)$	5	$3/75$	$680/19448$
$(17,9)$	4	$5/95$	$1122/24310$
$(19,7)$	6	$13/154$	$3591/50388$
$(19,9)$	5	$22/280$	$5510/92378$
$(19,10)$	4	$2/280$	$209/92378$
$(19,11)$	4	$35/228$	$10944/75582$
$(19,13)$	3	$3/86$	$570/27132$
$(23,7)$	8	$11/492$	$4807/245157$
$(23,8)$	7	$14/984$	$6072/490314$
$(23,9)$	7	$484/1630$	$237567/817190$
$(23,10)$	6	$187/2282$	$88803/1144066$
$(23,11)$	6	$2141/2694$	$1072260/1352078$
$(23,12)$	5	$345/2694$	$168291/1352078$
$(23,13)$	5	$1707/2282$	$853116/1144066$
$(23,14)$	4	$42/1630$	$19228/817190$
$(23,15)$	4	$396/984$	$193798/490314$
$(23,17)$	3	$5/207$	$2024/100947$

In each listed case the counting bound (6.1) *is satisfied.*

For $p \le 19$ the computation was done independently by two of the authors. The full search uses two reductions: it suffices to test one information set per $\mathrm{AGL}(1,p)$-orbit, and, for a fixed information set, one error pattern T per $\mathrm{AGL}(1,p)$-orbit (if $\sigma(T) \subseteq \mathscr{J}$ then $(\sigma\rho^{-1})(\rho(T)) \subseteq \mathscr{J}$). With these, $p = 23$ takes under a minute in Python. The code and complete output are available in the repository [Repo]; a short check of Example 7.7 is in Appendix B.

At $p = 23$, failure has become common: for $k = 11$ about 79% of all information sets fail, although some information set still works for every k.

Remark 7.6. The most natural choice, the "interval" information set $\mathscr{I} = \{0, 1, \ldots, k-1\}$, works for every k when $p \le 17$, but fails for $k \in \{7, 9, 11\}$ when $p = 19$ and for $k \in \{7, 9, 10, \ldots, 15\}$ when $p = 23$. So a good information set must be chosen with some care even for moderate p.

Example 7.7 (Failure despite the counting bound)**.** Let $p = 17$, $k = 9$, so $C = \mathrm{RS}(\mathbb{F}_{17}, 9)$ is a $[17, 9, 9]$ code with $t = 4$. Take check positions

$$\mathscr{J} = \{0, 1, 2, 3, 4, 7, 9, 10\} \subset \mathbb{F}_{17},$$

so that $\mathscr{I} = \mathbb{F}_{17} \setminus \mathscr{J}$. An enumeration of all 272 affine maps shows that none sends the error support $T = \{0, 1, 4, 5\}$ into $\mathscr{J}$. Thus $\mathrm{AGL}(1, 17)$ is not a 4-PD-set relative to $\mathscr{I}$, although

$$272\binom{8}{4} = 19040 > 2380 = \binom{17}{4}.$$

7.2 A counterexample to (C1) for every information set

Theorem 7.8. *Let $p = 37$ and $D = 32P_\infty$, so $C = \mathrm{RS}(\mathbb{F}_{37}, 17)$ with parameters $[37, 17, 21]$, $r = 20$ and $t = 10$. No subset of $\mathrm{PAut}(C) = \mathrm{AGL}(1, 37)$ is a full PD-set, relative to any information set. Hence conjecture (C1) is false.*

Proof. $|\mathrm{AGL}(1, 37)|\binom{20}{10} = 1332 \cdot 184756 = 246094992 < 348330136 = \binom{37}{10}$. Apply Theorem 6.3. □

Remark 7.9. A direct check shows that $p = 37$ is the smallest prime for which the counting bound, applied to $\mathrm{AGL}(1, p)$ with $s = t$, rules out a full PD-set in some dimension $2 \le k \le p - 2$. At $p = 37$ this happens exactly for $k \in \{17, \ldots, 25\} \cup \{27\}$, so $k = 17$ is the smallest case. For $29 \le p \le 31$ the counting bound decides nothing, and the exhaustive check is beyond a simple Python program; see [Repo] for what a compiled implementation needs.

8 Codes over $\mathbb{F}_{p^2}$

Assume $p \equiv 3 \pmod 4$ and $p \ge 7$. Put $E_2 = \sum_{P \in O_2} P$, so $n = 2p(p-1)$, and let $D = mP_\infty$ with $m < n$. The group H of (4.8) fixes D and E_2. By Proposition 4.6 it acts regularly on the n coordinates, so its image $\pi_{E_2}(H)$ has exactly n elements. The obstruction below applies to any regular group.

Theorem 8.1 (Regular groups correct few errors)**.** *Let C be an $[n, k, d]$ code with $k < n$, and let $K \le \mathrm{PAut}(C)$. If $1 \le s \le \min(t, n - k)$ and*

$$\left(\frac{n}{n-k}\right)^s > |K|,$$

then no subset of K is an s-PD-set, relative to any information set. In particular, if $|K| = n$ this holds for every $s > \log n / \log\big(n/(n-k)\big)$ with $s \le \min(t, n-k)$.

Proof. For $0 \le i < s \le n - k$ we have $(n-i)/(n-k-i) \ge n/(n-k)$, so

$$\binom{n}{s} \Big/ \binom{n-k}{s} = \prod_{i=0}^{s-1} \frac{n-i}{n-k-i} \ge \left(\frac{n}{n-k}\right)^s > |K|.$$

Apply Theorem 6.3. □

Corollary 8.2 (An infinite family of counterexamples to (C2))**.** *Let* $p \equiv 3 \pmod 4$, $p \geq 7$, $n = 2p(p-1)$, $g = (p-1)/2$ *and*

$$m = \tfrac{n}{2} + g - 1, \qquad C = C_X(mP_\infty, E_2).$$

Then C is an $[n, n/2, \geq n/2 - g + 1]$ code, with $t \geq \lfloor (p-1)(2p-1)/4 \rfloor$. For $s_0 = \lfloor \log_2 n \rfloor + 1$ we have $s_0 \leq t$, and $\pi_{E_2}(H)$ is not an s_0-PD-set, hence not a full PD-set, relative to any information set.

Proof. Since $2g - 1 \leq m < n$, the map ev_{E_2} is injective and $k = \ell(D) = m + 1 - g = n/2$. Also $d \geq n - m = n/2 - g + 1$, and $n/2 - g = (p-1)(2p-1)/2$, which gives the bound on t. Next, $s_0 \leq \log_2(2p^2) + 1 = 2 + 2\log_2 p$. The inequality $3 + 2\log_2 p \leq (p-1)(2p-1)/4$ holds at $p = 7$ (where $8.62 \leq 19.5$), and its right side grows faster for $p \geq 7$. Since $t > (p-1)(2p-1)/4 - 1$, this gives $s_0 \leq t$. Finally $(n/(n-k))^{s_0} = 2^{s_0} > n = |\pi_{E_2}(H)|$, and Theorem 8.1 applies. □

p	n	k	lower bound for t	s_0
7	84	42	19	7
11	220	110	52	8
19	684	342	166	10
23	1012	506	247	10
31	1860	930	457	11

Example 8.3 (The case $p = 7$)**.** For $p = 7$, Corollary 8.2 gives $m = 44$ and the $[84, 42, \geq 40]$ code $C_X(44P_\infty, E_2)$ over $\mathbb{F}_{49}$, with $t \geq 19$. Here

$$84\binom{42}{7} = 2266179552 < 4529365776 = \binom{84}{7},$$

while $s = 6$ is not excluded by the counting bound. So the stabilizer image is not a 7-PD-set.

Remark 8.4. These results concern the geometric stabilizer proposed in (C2), not the full group $\mathrm{PAut}(C)$, which we have not determined. The stabilizer of $D = mP_\infty$ in G_{p^2} is exactly H, so if $\mathrm{PAut}(C)$ is induced by automorphisms of X preserving D and E_2, then $\mathrm{PAut}(C) = \pi_{E_2}(H)$ and Corollary 8.2 applies to the full permutation group. The results of [W, JK06] that give such an identification assume n large compared with $\deg D$, and this fails for the half-rate family ($n < 2\deg D$). So the question remains open (Problem 10.3). In any case Theorem 8.1 shows that any group $K \leq \mathrm{PAut}(C)$ of polynomial size in n can support permutation decoding of only $O(\log n)$ errors at fixed rate.

9 Complexity and computation of the action

The conjectures of [Jo05] combined a group-size estimate with the claim that a group is a full PD-set. These must be separated. If syndromes are computed with a dense $r \times n$ matrix, one trial costs $O(nr)$ field operations plus $O(n)$ coordinate work. A worst-case scan of S therefore costs $O(|S|nr)$ field operations. The estimates $|\mathrm{AGL}(1,p)| = O(p^2)$ and $|H| = O(p^2)$ describe group

orders, not total costs, and they do not show asymptotic superiority over Reed–Solomon or AG decoders [HvLP, GS99]. The alphabet sizes p and p^2 also grow with n, which matters for bit complexity.

On the other hand, a general algorithm for the automorphism group of a linear code, such as Leon's [L], can be expensive. (The GUAVA command `PermutationAutomorphismGroup` took over a day for $p = 13$ on 2005 hardware [Jo05].) For the codes considered here, the formulas (4.4) and (4.8) give the action of G_p, H and G_{p^2} on coordinates directly, at negligible cost. This remains a practical advantage of geometric constructions.

10 Open problems

Problem 10.1 (Partial permutation decoding)**.** For $C_X(mP_\infty, E_2)$ over $\mathbb{F}_{p^2}$, determine the largest $s = s(p, m, \mathscr{I})$ for which $\pi_{E_2}(H)$ is an s-PD-set relative to $\mathscr{I}$. Compare it with the bound of Theorem 8.1 and with the value for the full group $\mathrm{PAut}(C)$, and characterize the best information sets.

Problem 10.2. For $\mathrm{RS}(\mathbb{F}_p, k)$ and $p \in \{29, 31\}$, determine for which k some information set makes $\mathrm{AGL}(1, p)$ a full PD-set. (By Proposition 7.5, such a set exists for every k when $p \leq 23$; by Theorem 7.8, not when $p = 37$, $k = 17$.) More generally, describe the information sets that fail, as in Proposition 7.5 and Example 7.7.

Problem 10.3. Determine $\mathrm{PAut}(C_X(mP_\infty, E_2))$, and decide whether it equals $\pi_{E_2}(G_{p^2})$, for the codes of Section 8.

Problem 10.4 (Burst errors)**.** The curve $x^p - x = y^2$ has the separated form studied in [CTO26], and [LLMS26] constructs burst-error PD-sets on related curves. Do the translations τ_b and the maps $(x, y) \mapsto (ux, \eta y)$ give PD-sets for bursts of errors supported on fibers of x or y in the codes $C_X(mP_\infty, E_2)$? How do such sets compare with the general bound of Theorem 8.1?

11 Summary

1. The $\mathbb{F}_p$ codes of length p are full-length Reed–Solomon codes, and their permutation group $\mathrm{AGL}(1, p)$ is larger than the geometric image (Theorem 7.1, Remark 7.2).

2. $\mathrm{AGL}(1, p)$ always gives 2-error permutation decoding (Theorem 7.3).

3. Full permutation decoding by $\mathrm{AGL}(1, p)$ depends on the information set from $p = 17$ on, though for $p \leq 23$ a good information set always exists (Proposition 7.5, Example 7.7), and fails for all information sets at $p = 37$ (Theorem 7.8).

4. Over $\mathbb{F}_{p^2}$, regular groups such as the geometric stabilizer correct only $O(\log n)$ errors by permutation decoding at fixed rate, giving infinitely many counterexamples to (C2) (Theorem 8.1, Corollary 8.2).

5. Equivariant AG codes are submodules of permutation modules, but the Riemann–Roch module need not be a permutation module (Example 5.3).

Acknowledgments

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This revision was prepared with the help of two AI systems, listed as authors at the first author's request. *ChatGPT (OpenAI)* prepared the first mathematical revision. It identified the $\mathbb{F}_p$ codes as Reed–Solomon codes, introduced the counting obstruction and the counterexamples at $p = 37$ and $p = 7$, proved the 2-PD result, gave the permutation-module counterexample and the proof of the orbit proposition, and corrected several statements of [Jo05]. In its response to the referee report, it found the $p = 17$ information set of Example 7.7 and the choice $m = n/2 + g - 1$ in Corollary 8.2. *Claude (Anthropic)* refereed that revision. It checked the proofs and numerical claims independently, including a brute-force determination of the permutation group of the $[7, 3, 5]$ code. It found errors in the GAP transcript, the lift criterion, the description of kernels, the group-action conventions, and the silent reinterpretation of point labels. It carried out the exhaustive computations of Proposition 7.5 (including $p = 23$), proposed the logarithmic obstruction of Theorem 8.1, verified the bibliography, located the recent literature [CTO26, LLMS26, CLLS], and prepared this version, including the background section, literature review, and glossary. Responsibility for the final content rests with the first author.

A Historical GUAVA session

The commands of [Jo05], which used GUAVA 2.0 [GUAVA], are kept for the record, with two typos corrected: `Length(P)` is now `Length(Pts)`, and the closing `if;` is now `fi;`. They are version dependent, and the results of this paper do not depend on them.

```
p:=7;; F:=GF(p);;
R:=PolynomialRing(F,2);;
indets:=IndeterminatesOfPolynomialRing(R);;
x:=indets[1];; y:=indets[2];;
Pts:=AffinePointsOnCurve(y^2-x^p+x,R,F);;
Length(Pts);
C:=OnePointAGCode(y^2-x^p+x,Pts,4,R);;
MinimumDistance(C);
G:=GeneratorMat(C);; PutStandardForm(G);; Display(G);
P:=PermutationAutomorphismGroup(C);; IdGroup(P);
Wt2:=ConstantWeightSubcode(WholeSpaceCode(p,GF(p)),2);;
for c in C do
  e:=Random(Wt2);; v:=PermutationDecode(C,c+e);;
  if c<>v then Print("\n Decoding fails. c,v = ",c," ",v,"\n"); fi;
od;
```

B Verification script

The following Python 3 code (which also runs in Sage) checks Example 7.7 and one entry of Proposition 7.5. Information sets are reduced modulo $\mathrm{AGL}(1,p)$, and subsets are stored as bit masks. The faster program `pd_affine.py` in [Repo], which also reduces error patterns modulo $\mathrm{AGL}(1,p)$, produces the whole table.

```
from itertools import combinations
p, J, T = 17, {0,1,2,3,4,7,9,10}, {0,1,4,5}
print([(u,b) for u in range(1,p) for b in range(p)
       if all((u*x+b) % p in J for x in T)])     # prints []

def failing_orbits(p, k):
    maps = [[(u*x+b) % p for x in range(p)]
            for u in range(1,p) for b in range(p)]
    full, t = (1 << p) - 1, (p-k)//2
    Ts = [sum(1 << i for i in S) for S in combinations(range(p), t)]
    seen, bad, tot = set(), 0, 0
    for I in combinations(range(p), k):
        if sum(1 << i for i in I) in seen: continue
        orb = {sum(1 << m[i] for i in I) for m in maps}
        seen |= orb; tot += 1
        Js = [full ^ o for o in orb]   # images of the check set
        if not all(any(s & ~Jm == 0 for Jm in Js) for s in Ts):
            bad += 1
    return bad, tot

print(failing_orbits(17, 9))                       # (5, 95)
```

Glossary of technical terms

Affine group $\mathrm{AGL}(1,q)$**.** The group of maps $x \mapsto ux+b$ on $\mathbb{F}_q$, $u \neq 0$; order $q(q-1)$; sharply 2-transitive.

AG code $C_X(D,E)$**.** The image of $L(D)$ under evaluation at the points of E.

Automorphism group $\mathrm{Aut}_F(X)$**.** The group of automorphisms of the curve X defined over F.

Borel subgroup. The stabilizer of ∞ in $\mathrm{PGL}_2(p)$; the upper-triangular classes.

Branch point. A point of $\mathbb{P}^1$ over which a cover is ramified; for $y^2 = x^p - x$, the points of $\mathbb{P}^1(\mathbb{F}_p)$.

Check positions $\mathscr{J}$**.** The coordinates complementary to an information set.

Counting bound. The necessary condition $|S|\binom{n-k}{s} \geq \binom{n}{s}$ for an s-PD-set S.

Divisor; degree. A formal integer combination of points; the weighted sum of its coefficients.

Equivariant (invariant) divisor. A divisor fixed by a group of automorphisms.

Evaluation map ev_E**.** $f \mapsto (f(P_1), \ldots, f(P_n))$.

Full PD-set. A t-PD-set, where $t = \lfloor (d-1)/2 \rfloor$.

Genus g**.** The basic invariant of a curve; here $g = (p-1)/2$.

Hamming weight $\mathrm{wt}(v)$**.** The number of nonzero coordinates of v.

Hasse–Weil bound. $||X(\mathbb{F}_q)| - q - 1| \leq 2g\sqrt{q}$.

Hyperelliptic curve; involution ι**.** A curve of genus ≥ 2 with a degree-two map to $\mathbb{P}^1$; the involution exchanging its sheets.

Information set $\mathscr{I}$**.** k coordinates on which projection of the code is bijective.

Large automorphism group. $|G| > |X(F)|$.

Maximal / minimal curve. A curve attaining the upper / lower Hasse–Weil bound.

MDS code. A code with $d = n - k + 1$; every k-set is an information set.

One-point code. An AG code with $D = mQ$ for a single point Q.

Orbit; stabilizer. $G \cdot \omega$ and $G_\omega = \{g : g\omega = \omega\}$; $|G \cdot \omega| = |G|/|G_\omega|$.

Permutation automorphism group $\mathrm{PAut}(C)$**.** Coordinate permutations preserving C.

Permutation decoding. Decoding by moving errors off an information set with code automorphisms and re-encoding.

Permutation module. A module with a basis permuted by the group.

$\mathrm{PGL}_2(p)$**,** $\mathrm{PSL}_2(p)$**.** The projective general and special linear groups over $\mathbb{F}_p$, of orders $p(p^2-1)$ and $p(p^2-1)/2$.

Ramification. A point of a cover is ramified if its stabilizer (inertia group) is nontrivial.

Reed–Solomon code $\mathrm{RS}(\mathbb{F}_q, k)$**.** Evaluations on $\mathbb{F}_q$ of polynomials of degree $< k$.

Regular action. Transitive with trivial stabilizers; then $|G|$ equals the number of points.

Riemann–Roch space $L(D)$**.** Functions f with $\mathrm{div}(f) + D \geq 0$, together with 0.

Semisimplification. The direct sum of the composition factors of a module.

Separates points. $L(D)$ separates $P \neq Q$ if some $f \in L(D)$ has $f(P) \neq f(Q)$.

Sharply 2-transitive. Exactly one group element maps any ordered pair of distinct points to any other.

s**-PD-set.** A set of automorphisms moving every set of at most s coordinates into the check positions.

Syndrome. $\mathsf{H}v^{\mathsf{T}}$ for a parity-check matrix H.

Systematic form. Generator matrix $(I_k \mid A)$ and parity-check matrix $(-A^{\mathsf{T}} \mid I_{n-k})$.

Very ample. A divisor whose complete linear system embeds the curve; on a curve, implied by $\deg D \geq 2g + 1$.

Weierstrass point. For a hyperelliptic curve, a fixed point of ι.

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